Homework 5

MATH 441: DIFFERENTIAL EQUATIONS

NAME: DATE:

(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

1. Exercise §3.5 #9. Find the general solution of the given differential equation

$$2y'' + 3y' + y = t^2 + 3\sin t$$

$y_c = 2r^2 + 3r + 1 = 0$

$(2r+1)(r+1) = 0$

$r \in \{-\frac{1}{2}, -1\}$

$\boxed{y_c = C_1 e^{-\frac{1}{2}t} + C_2 e^{-t}}$

$y_{p_1} = C_1 t^2 + C_2 t + C_3$

$y'_{p_1} = 2C_1 t + C_2$

$y''_{p_1} = 2C_1$

$4C_1 + 6C_1 t + 3C_2 + C_1 t^2 + C_2 t + C_3 = t^2$

$(C_1 - 1)t^2 + (6C_1 + C_2)t + 4C_1 + 3C_2 + C_3 = 0$

$C_1 = 1 \quad C_2 = -6 \quad C_3 = 14$

$\boxed{y_{p_1} = t^2 - 6t + 14}$

$y_{p_2} = C_1 \sin t + C_2 \cos t$

$y'_{p_2} = C_1 \cos t - C_2 \sin t$

$y''_{p_2} = -C_1 \sin t - C_2 \cos t$

$-2C_1 \sin t - 2C_2 \cos t + 3C_1 \cos t - 3C_2 \sin t + C_1 \sin t + C_2 \cos t = 3\sin t$

$\sin t(-2C_1 - 3C_2 + C_1 - 3) + \cos t(-2C_2 + 3C_1 + C_2) = 0$

$C_1 = -3C_2 - 3 \qquad 0 = -2C_2 - 9C_2 - 9 + C_2$

$C_1 = \frac{-3}{10} \qquad -\frac{9}{10} = C_2$

$\boxed{y_{p_2} = \frac{-3}{10}\sin t - \frac{9}{10}\cos t}$

$\boxed{y = C_1 e^{-\frac{1}{2}t} + C_2 e^{-t} + t^2 - 6t + 14 - \frac{3}{10}\sin t - \frac{9}{10}\cos t}$

2. Exercise §3.5 #17. Find the solution of the given intial value problem.

$$y'' - 2y' + y = te^t + 4, \qquad y(0) = 1, \qquad y'(0) = 1$$

$y_c = r^2 - 2r + 1 = 0$

$(r-1)^2 = 0$

$\boxed{y_c = C_1 e^t + C_2 t e^t}$

$y_{p_1} = (C_1 t e^t + C_2 e^t)t^2$

$= t^2 e^t (C_1 t + C_2)$

$(D-1)^2 y_{p_1} = te^t$

let $y_{p_1} = e^t v(t)$ for simplicity

$(D-1)y_{p_1} = e^t v(t) + e^t v'(t) - e^t v(t) = e^t v'(t)$

$(D-1)^2 y_{p_1} = (D-1)(D-1)y_{p_1} = (D-1)e^t v'(t) = e^t(v'(t) + v'(t) - v'(t) = e^t v''(t) = te^t$

$v''(t) = t$

$v(t) = \frac{1}{6}t^3$

$\boxed{y_{p_1} = e^t(\frac{1}{6}t^3)}$

$y_{p_2} = 4 \qquad 4 = 4$

$y = C_1 e^t + C_2 t e^t + \frac{1}{6}t^3 e^t + 4$

$y' = C_1 e^t + C_2 e^t + C_2 t e^t + \frac{1}{2}t^2 e^t + \frac{1}{6}t^3 e^t$

$y(0) = C_1 + 4 = 1 \qquad C_1 = -3$

$y'(0) = -3 + C_2 = 1 \qquad C_2 = 4$

$\boxed{y(t) = -3e^t + 4te^t + \frac{1}{6}t^3 e^t + 4}$

In each of Problems 3 and 4, use the method of variation of parameters to find a particular solution of the given differential equation. Then check your answer using the method of undetermined coefficients.

3. Exercise §3.6 #1.

$$y'' - 5y' + 6y = 2e^t$$

$y_c = r^2 - 5r + 6 = 0$

$(r-2)(r-3) = 0$

$r \in \{2, 3\}$

$\boxed{y_c = c_1 e^{2t} + c_2 e^{3t}}$ $y_1 = e^{2t}$ $y_2 = e^{3t}$

$y_p = u_1 y_1 + u_2 y_2$

$y_p' = u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2'$ set $u_1' y_1 + u_2' y_2 = 0$

$= u_1 y_1' + u_2 y_2'$

$y_p'' = u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2''$

$u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2'' - 5u_1 y_1' - 5u_2 y_2' + 6u_1 y_1 + 6u_2 y_2 = 2e^t$

$2u_1' e^{2t} + 4u_1 e^{2t} + 3u_2' e^{3t} + 9u_2 e^{3t} - 10u_1 e^{2t} - 15u_2 e^{3t} + 6u_1 e^{2t} + 6u_2 e^{3t} = 2e^t$

$e^{2t}(2u_1' + 4u_1 - 10u_1 + 6u_1) + e^{3t}(3u_2' + 9u_2 - 15u_2 + 6u_2) = 2e^t$

$2u_1' e^{2t} + 3u_2' e^{3t} = 2e^{-t}$

$\boxed{2u_1' e^{2t} + 3u_2' e^{3t} = 2e^t}$

$\boxed{u_1' e^{2t} + u_2' e^{3t} = 0}$

$-u_1' e^{2t} = 2e^t$

$u_1' = -2e^{-t}$

$\boxed{u_1 = 2e^{-t}}$

$u_2' = 2e^{-2t}$

$\boxed{u_2 = -e^{-2t}}$

$\boxed{y_p = 2e^t - e^t = e^t}$

check w/ undet. coeff. see page 5

4. Exercise §3.6 #3.

$$y'' + 2y' + y = 3e^{-t}$$

$y_c = r^2 + 2r + 1 = 0$

$(r+1)^2 = 0$

$r \in \{-1\}$

$y_c = c_1 e^{-t} + c_2 t e^{-t}$

$y_p = u_1 e^{-t} + u_2 t e^{-t}$

$y_p' = u_1' e^{-t} - u_1 e^{-t} + u_2' t e^{-t} + u_2 e^{-t} - u_2 t e^{-t}$ set $u_1' e^{-t} + u_2' t e^{-t} = 0$

$y_p' = -u_1 e^{-t} + u_2 e^{-t} - u_2 t e^{-t}$

$y_p'' = -u_1' e^{-t} + u_1 e^{-t} + u_2' e^{-t} - u_2 e^{-t} - u_2' t e^{-t} - u_2 e^{-t} + u_2 t e^{-t}$

$-u_1' e^{-t} + u_2' e^{-t} - u_2' t e^{-t} + u_1 e^{-t}(1-2+1) + u_2 e^{-t}(-1-1+2) + u_2 t e^{-t}(1-2+1) = 3e^{-t}$

$-u_1' e^{-t} + u_2' e^{-t} - u_2' t e^{-t} = 3e^{-t}$

$-u_1' e^{-t} + u_2' e^{-t}(1-t) = 3e^{-t}$

$-u_1' + u_2'(1-t) = 3$

$u_1' + u_2' t = 0$

$u_2' t + u_2' - u_2' t = 3$

$u_2' = 3$

$u_2 = 3t$

$u_1 = -\frac{3}{2}t^2$

$y_p = -\frac{3}{2}t^2 e^{-t} + 3t^2 e^{-t}$

$\boxed{y_p = \frac{3}{2}t^2 e^{-t}}$

Undet. Coeff. see page 4

In each of Problems 5 through 7, find the general solution of the given differential equations.

5. Exercise §4.2 #13.

$$2y''' - 4y'' - 2y' + 4y = 0$$

$y_c = 2r^3 - 4r^2 - 2r + 4 = 0$

$r^3 - 2r^2 - r + 2 = 0$

2	1	-2	-1	2
		2	0	-2
	1	0	-1	0

$(r-2)(r^2-1) = 0$

$r \in \{2, 1, -1\}$

$y_c = c_1 e^{-t} + c_2 e^{t} + c_3 e^{2t}$ $\quad y_p = 0$

$$y(t) = c_1 e^{-t} + c_2 e^{t} + c_3 e^{2t}$$

6. Exercise §4.2 #14.

$$y^{(4)} - 4y''' + 4y'' = 0$$

$r^4 - 4r^3 + 4r^2 = 0$

$r^2(r^2 - 4r + 4) = 0$

$r^2(r-2)^2 = 0$

$r \in \{0, 2\}$

$y_c = c_1 + c_2 t + c_3 e^{2t} + c_4 t e^{2t}$ $\quad y_p = 0$

$$y(t) = c_1 + c_2 t + c_3 e^{2t} + c_4 t e^{2t}$$

7. Exercise §4.3 #4.

$$y''' - y' = 2\sin t$$

$y_c = r^3 - r = 0$

$r(r^2-1) = 0$

$r \in \{0, 1, -1\}$

$y_c = c_1 + c_2 e^t + c_3 e^{-t}$

$y_p = c_1 \sin t + c_2 \cos t$

$y'_p = c_1 \cos t - c_2 \sin t$

$y''_p = -c_1 \sin t - c_2 \cos t$

$y'''_p = -c_1 \cos t + c_2 \sin t$

$-2c_1 \cos t + 2c_2 \sin t = 2\sin t$

$c_2 \sin t - c_1 \cos t = \sin t$

$c_2 = 1$

$c_1 = 0$

$y(t) = c_1 + c_2 e^t + c_3 e^{-t} + \cos t$

3) $y'' - 5y' + 6y = 2e^t$ $\quad y_c = c_1 e^{2t} + c_2 e^{3t}$

Unde. Coeff.:

$y_p = c_1 e^t$

$y_p' = c_1 e^t$

$y_p'' = c_1 e^t$

$c_1 e^t - 5c_1 e^t + 6c_1 e^t = 2e^t$

$c_1 e^t = e^t$

$c_1 = 1$

$\boxed{y_p = e^t}$

4) $y'' + 2y' + y = 3e^{-t}$ $\quad y_c = c_1 e^{-t} + c_2 t e^{-t}$

Unde. Coeff.: $y_p = (c_1 e^{-t}) t^2$

$y_p' = 2c_1 t e^{-t} - c_1 t^2 e^{-t}$

$y_p'' = 2c_1 e^{-t} - 2c_1 t e^{-t} - 2c_1 t e^{-t} + c_1 t^2 e^{-t}$

$e^{-t}(2c_1 - 4c_1 t + c_1 t^2 + 4c_1 t - 2c_1 t^2 + c_1 t^2) = 3e^{-t}$

$2c_1 = 3$

$c_1 = \frac{3}{2}$

$\boxed{y_p = \frac{3}{2} t^2 e^{-t}}$